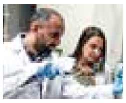
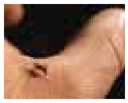


Insulin in a pill
Researchers have developed an effective method to administer insulin orally.
http://dgit.in/may21-33



Modifying mosquito genetics
Mosquito gut genes can be altered to make them spread antimalarial genes to the next generation. http://dgit.in/may21-34

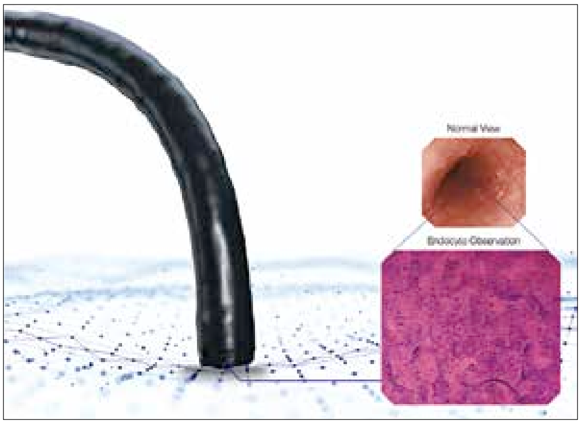
and EndoBRAINEYE, support software that is capable of identifying the problematic polyps.

The way this benefits the patient is that the endoscopic machine provides an automatic analysis, and prevents the need for collection of samples to be sent away for testing.

We spoke to Dr. Mohan Ramchandani, Director of Interventional Endoscopy at AIG Hospitals to understand more about the initiative and how AI diagnosis can help improve local healthcare, "AI diagnostic systems in colonoscopy will greatly improve the quality of endoscopic detection and diagnosis of colonic polyps, which if detected early can greatly improve the survival rates of patients with colon cancer. Colonoscopy is the gold standard diagnostic procedure for colon cancer detection, however early detection makes the difference which is still not achieved in countries like India considering the high population and less number of gastroenterologists available per capita. AI diagnostic systems in colonoscopy will improve the quality of detection and diagnosis even in remote places where the number of gastroenterologists are lower compared to metropolises."

We asked about the challenges faced by endoscopists in India and the benefit offered by AI, Dr. Ramchandani replied, "Current challenges in endoscopy include the lack of screening practices in place unlike in western countries, where colonoscopy is a standard screening procedure after a certain age. In the absence of preventive colonoscopy screening practices and limited awareness among people, colon cancers are generally detected at a later stage which otherwise could be detected early. Colon cancer is a kind of cancer that can be prevented if detected early. Artificial intelligence in colonoscopy will greatly improve the polyp detection rate by assisting in the detection of small and flat polyps which could be missed by human beings. In addition to the detection of polyps, AI will help the endoscopist in performing optical biopsies and make a confirmed diagnosis which otherwise requires a small piece to be cut and sent to a pathology lab for confirmation. Hence, AI will increase the Polyp and Adenoma detection rates by providing a reliable diagnosis which would lead to an increase in survival rates."

As part of the initiative, a virtual seminar on Olympus Endoscopic AI Diagnosis Education in India was held in March this year (it was a Zoom call). During that conference, Dr. Goutham Reddy Katukuri, Consultant of Gastroenterology of AIG Hospitals said,



The magnification offered by Endocyto

"Visual diagnosis of colorectal polyps may improve the cost-effectiveness of colonoscopy and reduce complications of polypectomy. A more precise diagnosis can be expected in the colonoscopic inspection using the AI diagnostic support system, and the difference in the diagnosis among observers may be corrected."

During the conference, Dr. D Nageshwar Reddy, Chairman and Chief of Gastroenterology at the Asian Institute of Gastroenterology probed Dr. Yuichi Mori, Clinical Effectiveness Research Group at University of Oslo in Norway, about the legal implications of the diagnosis made using AI, who should bear the responsibility of the diagnosis, the machine or the doctor. Dr. Mori replied, "This is very important discussion. Unfortunately, we do not have a clear answer yet. Regarding the diagnosis, it is thought that the location of legal responsibility lies with the doctors at present, however, it is the matter that is currently being discussed from various angles, such as if the accuracy of AI becomes 99% or more in the future (if it is better than the doctors), whether the location of responsibility should remain with the doctors or not."

Towards the end of the meet, we got some indications of future directions of the project. Mr. Satoshi Hemmi, Ministry of Internal Affairs and Communications, Japan said, "In order for new technologies, particularly AI, to be utilized in various countries, we believe it is necessary to customize them in accordance with the lives of people living in those regions." Mr. Hemmi went on to add, "I feel that this project has created a strong connection between doctors of India and Japan. Regardless of the medical field, it becomes an asset as a bond between Japan and India. I hope that this will trigger more and more collaboration among healthcare ICT and improve the level of healthcare in both countries."

We asked Dr. Ramchandani how the collaboration with Japan helped India, "AIG has already been leading the research, education, and healthcare in the field of gastroenterology in India and is one of the best endoscopy super speciality centers in the world. Collaborating with Japanese doctors in the field of AI for colonoscopy helped to bring this technology to India and get the required initial training. As a result of this collaboration, AI based endoscopic procedures for early detection and diagnosis will be made possible and AIG will lead the initiative in India to train other gastroenterologists in this. field. Eventually, AI endoscopy will help the gastroenterologists to provide better patient care."